

```
#Step 1 : Import Libraries
import pandas as pd
```

```
#Step 2 : Read / Load data
df = pd.read_csv('https://raw.githubusercontent.com/YBIFoundation/Dataset/main/Salary%20Data.csv')
```

```
df.head() #returns first 5 rows
```

	Experience Years	Salary
0	1.1	39343
1	1.2	42774
2	1.3	46205
3	1.5	37731
4	2.0	43525

Next steps: [Generate code with df](#) [New interactive sheet](#)

```
df.info() #shows information about file
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 40 entries, 0 to 39
Data columns (total 2 columns):
#   Column          Non-Null Count  Dtype
---  -
0   Experience Years  40 non-null    float64
1   Salary           40 non-null    int64
dtypes: float64(1), int64(1)
memory usage: 772.0 bytes
```

```
df.describe() #shows summary statistics
```

	Experience Years	Salary
count	40.000000	40.000000
mean	5.152500	74743.625000
std	2.663715	25947.122885
min	1.100000	37731.000000
25%	3.200000	56878.250000
50%	4.600000	64472.500000
75%	6.875000	95023.250000
max	10.500000	122391.000000

```
#Step 3 : Define y(output) and x(input)
df.columns
```

```
Index(['Experience Years', 'Salary'], dtype='object')
```

```
y = df['Salary']
x = df[['Experience Years']]
```

```
y
```

```
      Salary
0    39343
1    42774
2    46205
3    37731
4    43525
5    39891
6    48266
7    56642
8    60150
9    54445
10   64445
11   60000
12   57189
13   60200
14   63218
15   55794
16   56957
17   57081
18   59095
19   61111
20   64500
21   67938
22   66029
23   83088
24   82200
25   81363
26   93940
27   91000
28   90000
29   91738
30   98273
31  101302
32  113812
33  111620
34  109431
35  105582
36  116969
37  112635
38  122391
39  121872

dtype: int64
```

```
x
```

Experience Years	
0	1.1
1	1.2
2	1.3
3	1.5
4	2.0
5	2.2
6	2.5
7	2.9
8	3.0
9	3.2
10	3.2
11	3.5
12	3.7
13	3.8
14	3.9
15	4.0
16	4.0
17	4.1
18	4.3
19	4.5
20	4.7
21	4.9
22	5.1
23	5.3
24	5.5
25	5.9
26	6.0
27	6.2
28	6.5
29	6.8
30	7.1
31	7.9
32	8.2
33	8.5
34	8.7
35	9.0
36	9.5
37	9.6
38	10.3
39	10.5

Next steps:

[Generate code with x](#)[New interactive sheet](#)

```
#Step 4 : Train Test split
from sklearn.model_selection import train_test_split
```

```
x_train , x_test , y_train , y_test = train_test_split(x , y , random_state = 2529)
```

```
x.shape , x_train.shape , x_test.shape
```

```
((40, 1), (30, 1), (10, 1))
```

```
y.shape , y_train.shape , y_test.shape
```

```
((40,), (30,), (10,))
```

```
#Step 5 : Select a model  
from sklearn.linear_model import LinearRegression  
model = LinearRegression()
```

```
#Step 6 : Train model  
model.fit(x_train , y_train)
```

```
▼ LinearRegression ⓘ ?  
LinearRegression()
```

```
model.intercept_
```

```
np.float64(27010.963630394683)
```

```
model.coef_
```

```
array([9343.3671387])
```

```
#Step 7 : Prediction  
y_pred = model.predict(x_test)
```

```
#Step 8 : Evaluate Performance  
from sklearn.metrics import mean_absolute_percentage_error
```

```
mean_absolute_percentage_error(y_test , y_pred)
```

```
0.06230870755908321
```

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